E16

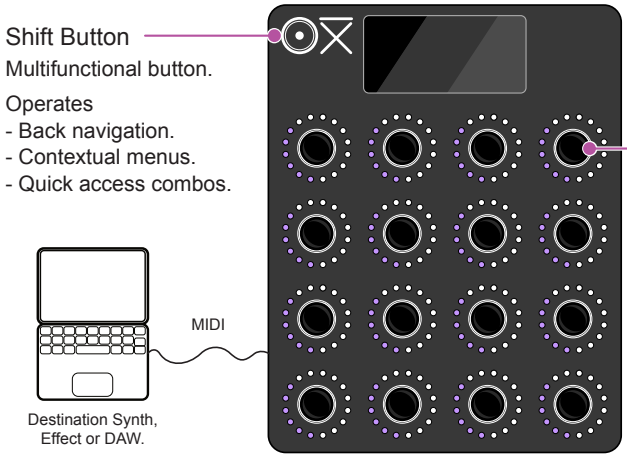
Advanced MIDI Parameter Controller

Version 1.3.0 a

Hardware Overview

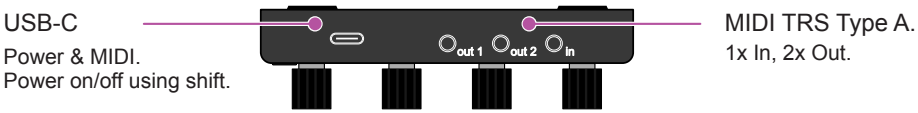
The E16 is a compact MIDI parameter controller which can communicate MIDI over TRS, USB or Bluetooth connections. There are 16 push encoders immediately available on the panel which can be configured for control of a destination device's MIDI CC, Control Change, PC, note and other MIDI parameters. The controls are organised in Scenes. A scene contains 12 pages, each page with 16 parameter controls. Each parameter can have up to 2 controlled destinations and 1 push action. Parameter values at any given time can be stored as a preset and recalled later. So a scene can hold 192 parameters, 384 destinations. Navigation is performed using the 16 push / encoder knobs and the shift button. The E16 can also host and control Lua scripts.

Front Panel



Important: Encoder Type.
The encoder type setting found in the system config must be set to match your hardware encoders in order for the software to manage this behavior. The options are normal (non-detented) and detented.

Rear / Edge



The E16 is supplied with this quick start guide, USB-C to A cable, and a high quality travel case. The full manual is available for download from the OXI Instruments site at oxiinstruments.com. The firmware and settings can also be managed by connecting the E16 to the OXI App.

Power On / Off

Power is supplied through the USB-C connection. Connect to a USB power supply, computer or power bank. It is recommended to use a supply greater that 200mA. The multifunctional button embedded in the OXI 'O' is used for turning the device on and off. This is called the shift button. To turn off, navigate to the main, top level scenes menu. Tap the shift button to open config and power off.



Power On
Tap the [Shift] button



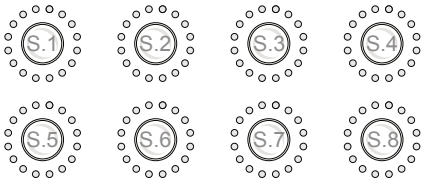


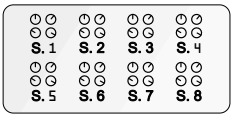
Power Off
In the home scene page,
Tap the [Shift] button, then hold the red lit encoder.

Navigating E16

The main 'home' screen location in the structure of OXI **E16** is presented when first powering on. This display will show the 8 available scenes labelled S.1 to S.8. The encoder knobs will perform selection of each scene. Any knob which is controlling a menu command will be lit. Tap a knob to select the option. Tap Knobs 9-16 to access the other 8 scenes, 9-16 and tap [Shift] for the configuration menu.

Main Menu: Scene & System Selection





Up to 8 Scenes are immediately accessible.

Tap [Shift] for config and on/off options.

In several screens, the knob position matches the position of the displayed elements. For example the scenes presented from the home page match the 1-8 knob positions.

Conventions



Turn
Edit or change value of the the respective parameter

The rotary LEDs will indicate a parameter value setting.



Tap
Select the respective displayed option. Example, to open a scene.

Any knob which has a currently selectable menu option will be lit.



[Shift] + Tap
Access secondary options and menu commands.



Tap [Shift]
Navigate backup in the menu hierarchy structure

The [Shift] button is used as a classic shift, to access secondary parameters or to backup or exit menus. Shift also opens the 'Hold' menu options in the control view.

Scene Organisation

A scene is a function to help organise the controls and saves all of the E16 settings. The scene options are found in the main, home screen, presented on power up. To start, select a scene to work with or to setup. Within the control view one of the 16 parameters can also be selected. You can open one of the 8 direct access scenes from the main, home menu. Tap knob 9-16 to view and select the other 8 scenes.

Choose a Scene.
Tap knob to select a scene from the home screen. Tap knob 9-16 to switch scene selection.



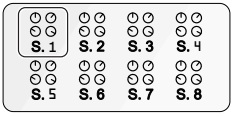
View or Change Controls
Tap [Shift] to toggle between the page selection/ alt menu or the control view.



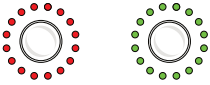
Page or Menu Option
Tap a knob 1-12 to change the page group or 13-16 to choose an Alt menu option.



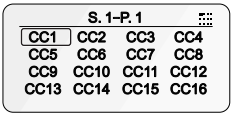
Home / Scenes



This is the main, home screen and is available upon power up. If any scene parameters have been edited, exiting from the page menu to the home screen will prompt to save the scene.



Control View

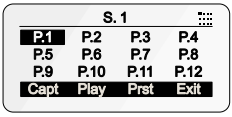


Turn a knob to adjust the MIDI output value for the respective parameter.





Alt Menu



Default Alt Menu Options: Pages +

Capt - Looper Record. Arms parameter automation. Will capture the control value changes while turning any knob. Tap to end.

Play. Sets the Play mode to On/Play or Off/ Stop. Determines if loops play or not.

Prst - Preset. Opens the load and save preset management menu.

Exit. Exit the and backup from the scene and back to the home scenes page.

Scene Presets

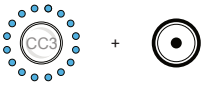
Scenes can be saved to the E16 memory. Each scene contains 16 presets which stores the parameter values of all the knobs at a given time. Presets can be individually named and can be recalled for later use.

Different configurations are therefore possible in order to match various hardware setups. The preset option is located in Alt menu, accessed from the control view by tapping [Shift] then pressing Knob 15 (Prst). The preset management menu is then navigated using the Knob which is lit. A maximum of 16 preset slots are available.

Parameter Setup

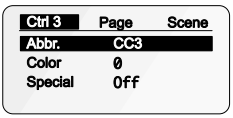
The workflow to navigate and setup each parameter in the scenes control view aims for quick and easy editing. The destination, behavior and name for each control can be edited.

Directly edit a control.
In the control view, hold a knob + tap [Shift]



Knobs represent each parameter control. The default knob assignment is in a numerical CC order. This can be changed.

The control change options can be edited for each individual parameter. This sets how the parameter is displayed on the E16 and how it affects the destination.



The 'Page' and 'Scene' settings are also accessible in this menu. Hold [Shift] + Tap Knobs to switch tabs for the editor focus. Tap a knob to select the control or page to edit from within these menus.

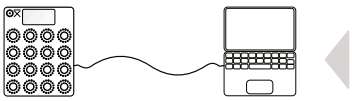
Parameter Options — Turn knob to navigate, tap knob to change.

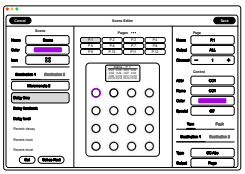
	Function	Description
Abbr	Label Abbreviation	Allows editing of up to 4 characters. This is the label used on the control page for the parameter. Editing of the label is made using the first 4 knobs, instructions as displayed.
Color	LED ring color	Assigns the color for the LED indicators for destination. Displayed around the specific control encoder knob to represent its value. If swap is active then colors can be set for each destination.
Special	Encoder Options	Can be set to Off or a special function. Snapshot Morph will capture and store an A and B value to assign as the knob range. Looper Capture will allow automation to be recorded by the knob when in the control view. Looper Playback, a recording can be controlled by pressing the knob. Swap Destination will enable switching the knobs control between destination 1 or 2 when pushing the knob. Random will set the value of parameters on the page to a random value. Pitch Bend Wheel sets the control to a MIDI pitch bend controller..
Dest1	Destination 1	Sets the parameters associated with controlling the destination device. Two device destinations can be assigned to each parameter. The options available are identical. Options includes range, type, target, output port, channel, etc
Dest2	Destination 2	
Push	Knob push behavior	Sets the behavior of the push command on the encoder and its associated display.
Copy	Copy Settings	Allows copying of these settings to be copied to another parameter.
Paste	Paste Settings	Allows pasting of previously copied settings to this parameter.
Erase	Erase settings	Allows clearing of the settings to defaults.

OXI App

While the setting up and configuration of the E16 is possible from the on-board controls, it is easier and quicker to create configurations using the OXI App and its integrated E16 Scene editor.

Download the App from the OXI site.

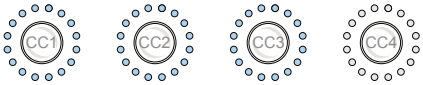




Groups

Each of the individual parameters can also be added to a ‘group’. A group is a collection of 2 or more parameters that can be controlled together as a collective set. Push and hold an encoder knob to view the group status. A maximum of 4 groups are available.

Set groups In the control view.



Group 1			
CC1	CC2	CC3	CC4
CC5	CC6	CC7	CC8
CC9	CC10	CC11	CC12
CC13	CC14	CC15	CC16

Tap the knobs to add to the group while holding the first knob. Each tap cycles through the behavior mode for the parameter. Example: Add knob encoder 2 and 3 to Group 1 controlled by knob 1.

Operating Behavior

Parameters displayed with a ‘solid’ bounding box will follow any changes made by a primary lead control, but can also be adjusted independently. The primary lead controls are shown with a ‘dotted’ boundary box. In the example above only CC1, when adjusted, will affect its own value plus the value of the other 2 parameters. When changing the CC2 or CC3 knob, only its own individual value is changed. Other parameters in the group are not affected.

The amount of control applied by the primary lead to a follower can be set for each follower parameter. The amount is set on each follower by holding & turning its knob. In this example hold + turn knob 2 (CC2) to set the amount and repeat for CC3.

Hold Knob + Tap the knob first time



Parameters displayed with a ‘dotted’ bounding box will act as a primary lead control. If any of these are adjusted, the values of the other parameters in the same group are also adjusted in harmony. The value change is relative to each of the parameter’s original value. In the example below, all three knobs will affect the value of itself and the other two parameters.

Hold Knob + Tap knob a second time.



Looper Automation

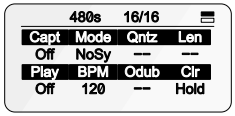
Recording is initiated by selecting capture mode to arm recording. 'Capt' can be selected using the Hold Menu Looper Capture . The Alt menu can also be configured to arm recording. Tap [Shift] to switch between control and alt menu view. The Hold menu is in the control view, open with [Shift]. Knobs can also be configured with a special setting to control recording commands. Recording mode determines synchronization. This can be set to No Sync which ignores a clock, Free Sync, quantized to clock, Fixed Sync, record to a set length.

Arm Capture & Record. Control view. Hold [Shift] + Tap Knob 1 'Capt', then tap or adjust a parameter knob to start it recording.

Stop Loop Recording. Control view, Tap a knob to stop it loop recording.

End Capture. Control view. Hold [Shift] + Tap Knob 1 'End' to disarm capture mode and stop recording.

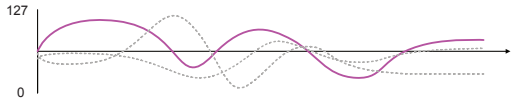
Play. Automation will playback based on whether the 'Play' option is set. Hold [Shift] + Tap (Knob 5) to start or stop playback.



In control view, turn the knob to record



Turn Parameter



Erase Automation. Control view. Turn a knob while capture is disarmed and playing normally to erase its automation. Alternatively, Hold [Shift] + Tap (Knob 8) 'Clr' to erase all automation.